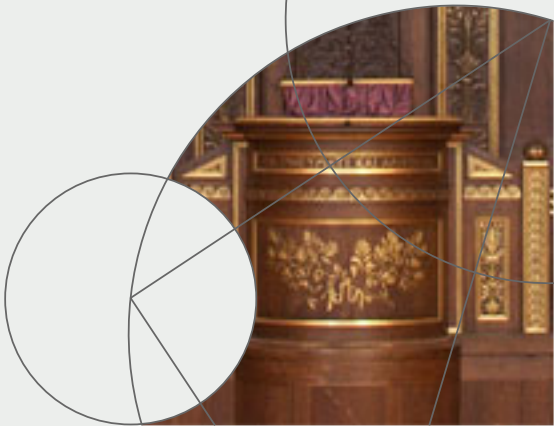
STEVENS INSTITUTE OF TECHNOLOGY

Data Acquisition and Processing I: Big Data

Rensheng Wang,
rwang1@stevens.edu
Dept. of Electrical and Computer Engineering
Stevens Institute of Technology



Time Series Analysis

- Time series analysis is the particular branch of statistics that studies processes evolving in time. All of statistics, like all of science, is about predicting the future from the past.
- ☞ If I measure the heights of 100 people in a town, I can construct a model for the distribution of the heights, and then before I measure the 101st person, I can make various probabilistic statements about what I expect his height to be.
- The difference in time series analysis is that the data points are explicitly ordered in time.
- Correlation is a most fundamental measurement used in the time series analysis.

Stationarity

- The foundation of time series analysis is stationarity.
- A time series $\{X_t\}$ is said to be strictly stationary if the joint distribution of $\{X_{t_1} \cdots, X_{t_k}\}$ is identical to that of $\{X_{t_1+l} \cdots, X_{t_k+l}\}$,

$$f(X_{t_1} \cdots, X_{t_k}) = f(X_{t_1+l} \cdots, X_{t_k+l}), \quad \text{for all } l$$

  where k is an arbitrary positive integer and $(t_1, \cdots, t_k)$ is a collection of k positive integers.
- In other words, strict stationarity requires that the joint distribution of $\{X_{t_1} \cdots, X_{t_k}\}$ is **invariant under time shift**.
- This is a very strong condition that is hard to verify empirically. A weaker version of stationarity is often assumed.
- A time series $\{X_t\}$ is weakly stationary if both the mean of $\{X_t\}$ and the covariance between $\{X_t\}$ and $\{X_{t-l}\}$ are time invariant, where l is an arbitrary integer.

Auto-Correlation

- Refer to the aforementioned correlation formula, and assign $X \to X_t$ and $Y \to X_{t-l}$. Correlation is reduced to Auto-Correlation.
- Let us define the mean and auto covariance for time series X_t at lag l

$$\mu = E(X_t); \qquad v_l = \mathrm{Cov}(X_t, X_{t-l}) = E[(X_t - \mu)(X_{t-l} - \mu)]$$

- The zero-order covariance $v_0 = \mathrm{Var}(X_t)$ is the mean-square size of the individual terms and the autocorrelation is

$$r_l = \frac{v_l}{v_0}$$

which has $r_0 = 1$ and $|r_l| \leq 1$. In addition, it is easy to see $r_l = r_{-l}$.

Meaning of Auto-Correlation

☞ What is the meaning of auto-correlation?

- Correlation of r_l between X_t and X_{t-l} tells you how much you learn about the value of X_t by knowing the value of X_{t-l}, without knowing anything about any other values.
- In general, suppose that ξ and η are two random variables with $E(\xi) = E(\eta) = 0$, $E(\xi^2) = \sigma_\xi^2, E(\eta^2) = \sigma_\eta^2$ and $E(\xi\eta) = \rho\sigma_\xi\sigma_\eta$.
- If we have no information about the value of ξ, our best predictor for the value η would be simply $\eta = 0$.
- To predict the value of η given ξ, we make a linear model $\eta = \beta\xi$.

Meaning of Auto-Correlation

- To predict the value of η given ξ, we make a linear model $\eta = \beta\xi$.
- We choose the value of β by minimizing the squared error:

$$\begin{aligned}
&\min_{\beta} E\left[(\eta - \beta\xi)^2\right] \\
&= \min_{\beta} E\left[\eta^2 - 2\beta\eta\xi + \beta^2\xi^2\right] \\
&= \min_{\beta} \left[E(\eta^2) - 2E(\beta\eta\xi) + E(\beta^2\xi^2)\right] \\
&= \min_{\beta} \left(\sigma_\eta^2 - 2\beta\rho\sigma_\eta\sigma_\xi + \beta^2\sigma_\xi^2\right)
\end{aligned}$$

- When β is chosen as $\beta = \rho\dfrac{\sigma_\eta}{\sigma_\xi}$, the above expression is minimized.
- Applying this with $\eta = X_t - \mu$, and $\xi = X_{t-l} - \mu$, both of zero mean and with equal variances, the best predictor for X_t given X_{t-l} is

$$X_t - \mu \sim r_l(X_{t-l} - \mu)$$

- The error in this prediction will be less than the zero-information prediction $X_t = \mu$.

Auto-Correlation Function (ACF) and Auto-Regressive (AR)

- Consider a weakly stationary time series X_t. When the linear dependence between X_t and its past values X_{t-l} is of interest, the concept of correlation is generalized to autocorrelation.
- The correlation factor between X_t and X_{t-l} is called the lag-l autocorrelation function (ACF) of X_t.
- If a time series X_t is stationary and can be explained by linear combinations of the past values X_{t-l} and some residual noise term, then the linear model is referred to as an autoregressive (AR) model.
- For example,

$$X_t = a_0 + a_1 X_{t-1} + \cdots a_p X_{t-p} + \epsilon_t$$

where ϵ_t is a random noise term, a_i are the linear coefficients with the past values and the order p is referred as AR(p) model.

Properties of AR(1) Models

For effective use of AR models, it pays to study their basic properties. We discuss properties of AR(1) and AR(2) models in detail and give the results for the general AR(p) model.

- AR(1) Model:

$$X_t = a_0 + a_1 X_{t-1} + \epsilon_t \tag{1}$$

Assuming that the series is weakly stationary, we have $E(X_t) = \mu$, $\mathrm{Var}(X_t) = v_0$, and $\mathrm{Cov}(X_t, X_{t-j}) = v_j$, where μ and v_0 are constant and v_j is a function of j, not t. We can easily obtain the mean, variance, and autocorrelations of the series as follows.

Taking the expectation of Eq. (1) and because noise term $E(\epsilon_t) = 0$, we obtain

$$E(X_t) = a_0 + a_1 E(X_{t-1})$$

Under the stationarity condition, $E(X_t) = E(X_{t-1}) = \mu$ and hence

$$\mu = a_0 + a_1 \mu \quad \text{or} \quad E(X_t) = \mu = \frac{a_0}{1 - a_1}$$

Properties of AR(1) Models

$$\mu = a_0 + a_1\mu \quad \text{or} \quad E(X_t) = \mu = \frac{a_0}{1-a_1}$$

- This result has two implications for X_t .
  First, the mean of X_t exists if $a_1 \neq 1$.
  Second, the mean of X_t is zero if and only if $a_0 = 0$.
- Thus, for a stationary AR(1) process, the constant term a_0 is related to the mean of X_t via $a_0 = (1-a_1)\mu$ and $a_0 = 0$ implies that $E(X_t) = 0$.
- Next, using $a_0 = (1-a_1)\mu$, the AR(1) model can be rewritten as

$$X_t - \mu = a_1(X_{t-1} - \mu) + \epsilon_t \tag{2}$$

- By repeated substitutions, the prior equation implies that

$$\begin{aligned} X_t - \mu =& \epsilon_t + a_1\epsilon_{t-1} + a_1^2\epsilon_{t-2} + \cdots \\ =& \sum_{i=0}^{\infty} a_1^i \epsilon_{t-i} \end{aligned}$$

Properties of AR(1) Models

- Thus, $X_t - \mu$ is a linear function of ϵ_{t-i} for $i \geq 0$. Using this property and the independence of the series $\{\epsilon_t\}$, we obtain $E[(X_t - \mu)\epsilon_{t+1}] = 0$.
- By the stationarity assumption, we have $\mathrm{Cov}(X_{t-1}, \epsilon_t) = E[(X_{t-1} - \mu)\epsilon_t] = 0$.
- This latter result can also be seen from the fact that X_{t-1} occurred before time t and ϵ_t does not depend on any past information.
- Taking the square, then the expectation of Eq. (2), we obtain

$$\mathrm{Var}(X_t) = a_1^2 \mathrm{Var}(X_{t-1}) + \sigma_\epsilon^2$$

  where σ_ϵ^2 is the variance of ϵ_t , and we make use of the fact that the covariance between X_{t-1} and ϵ_t is zero.
- Under the stationarity assumption, $\mathrm{Var}(X_t) = \mathrm{Var}(X_{t-1})$, so that

$$\mathrm{Var}(X_t) = \frac{\sigma_\epsilon^2}{1 - a_1^2}$$

  provided that $a_1^2 < 1$ (variance cannot be negative).

Autocorrelation Function of an AR(1) Model

- The requirement of $a_1^2 < 1$ results from the fact that the variance of a random variable is bounded and nonnegative.
- Consequently, the weak stationarity of an AR(1) model implies that $-1 < a_1 < 1$, that is, $|a_1| < 1$.
- In summary, the necessary and sufficient condition for AR(1) weakly stationary is $|a_1| < 1$.
- Multiplying Eq. (2) by ϵ_t , using the independence between ϵ_t and X_{t-1} , and taking expectation, we obtain

$$E[\epsilon_t(X_t - \mu)] = a_1 E[\epsilon_t(X_{t-1} - \mu)] + E(\epsilon_t^2) = E(\epsilon_t^2) = \sigma_\epsilon^2$$

where σ_ϵ^2 is the variance of ϵ_t .

Autocorrelation Function of an AR(1) Model

- Multiplying Eq. (2) by $X_{t-l}-\mu$,

$$(X_{t-l}-\mu)(X_t-\mu) = a_1(X_{t-1}-\mu)(X_{t-l}-\mu) + \epsilon_t(X_{t-l}-\mu)$$

- Taking expectation, and using the prior result, we have (recall that $\mathrm{Cov}(X_t, X_{t-l}) = v_l$)

$$E\{(X_{t-l}-\mu)(X_t-\mu)\} = a_1 E\{(X_{t-1}-\mu)(X_{t-l}-\mu)\} + E\{\epsilon_t(X_{t-l}-\mu)\}$$

$$\mathrm{Cov}(X_t, X_{t-l}) = a_1\,\mathrm{Cov}(X_{t-1}, X_{t-l}) + E\{\epsilon_t(X_{t-l}-\mu)\}$$

$$v_l = a_1 v_{l-1} + E\{\epsilon_t(X_{t-l}-\mu)\}$$

- When $l>0$, the noise term ϵ_t has nothing to do with the previous time series $X_{t-l}-\mu$

$$E\{\epsilon_t(X_{t-l}-\mu)\} = 0$$

- When $l=0$, see the results in the previous slide $E[\epsilon_t(X_t-\mu)] = \sigma_\epsilon^2$
- Put all together

$$v_l = \begin{cases} a_1 v_1 + \sigma_\epsilon^2 & \text{if } l=0 \\ a_1 v_{l-1} & \text{if } l>0 \end{cases}$$

Autocorrelation Function of an AR(1) Model

- For $l = 0$, $v_0 = a_1 v_1 + \sigma_\epsilon^2$
- For $l = 1$, $v_1 = a_1 v_0$
- v_1 in expression v_0 can be plugged in ($v_1 = a_1 v_0$), i.e.,

$$v_0 = a_1 v_1 + \sigma_\epsilon^2 = a_1^2 v_0 + \sigma_\epsilon^2$$

.

- Consequently, for a weakly stationary AR(1) model, we have

$$\mathrm{Var}(X_t) = v_0 = \frac{\sigma_\epsilon^2}{1 - a_1^2}$$

$$\mathrm{Cov}(X_t, X_{t-l}) = v_l = a_1 v_{l-1}, \quad \text{for } l > 0$$

- For the latter result, the ACF of X_t satisfies

$$r_l = \frac{v_l}{v_0} = a_1 r_{l-1} \quad \text{for} \quad l > 0$$

- Because $r_0 = 1$, we have $r_l = a_1^l$. This result says that the ACF of a weakly stationary AR(1) series decays exponentially with rate a_1 and starting value $r_0 = 1$.

Properties of AR(2) Models

- AR(2) Model:

$$X_t = a_0 + a_1 X_{t-1} + a_2 X_{t-2} + \epsilon_t \tag{3}$$

- Using the same technique as that of the AR(1) case, we obtain

$$E(X_t) = \mu = \frac{a_0}{1 - a_1 - a_2}$$

provided that $a_1 + a_2 \neq 1$.

- Using $a_0 = (1 - a_1 - a_2)\mu$, we can rewrite the AR(2) model as

$$(X_t - \mu) = a_1(X_{t-1} - \mu) + a_2(X_{t-2} - \mu) + \epsilon_t$$

- Multiplying the prior equation by $(X_{t-l} - \mu)$, we have

$$(X_{t-l} - \mu)(X_t - \mu) = a_1(X_{t-l} - \mu)(X_{t-1} - \mu) + a_2(X_{t-l} - \mu)(X_{t-2} - \mu) + (X_{t-l} - \mu)\epsilon_t.$$

- Taking expectation and using $E[(X_{t-l} - \mu)\epsilon_t] = 0$ for $l > 0$, we obtain

$$v_l = a_1 v_{l-1} + a_2 v_{l-2}, \quad \text{for} \quad l > 0.$$

Properties of AR(2) Models

- Dividing the above equation by v_0, we have the property ï£ij

$$r_l = a_1 r_{l-1} + a_2 r_{l-2}, \quad \text{for} \quad l > 0$$

for the ACF of X_t .

- In particular, the lag-1 ACF satisfies

$$r_1 = a_1 r_0 + a_2 r_{-1} = a_1 + a_2 r_1.$$

Therefore, for a stationary AR(2) series X_t , we have $r_0 = 1$,

$$r_1 = \frac{a_1}{1 - a_2}$$

$$r_l = a_1 r_{l-1} + a_2 r_{l-2}, \quad l \geq 2$$

- The above result says that the ACF of a stationary AR(2) series satisfies the second-order difference equation

$$(1 - a_1 B - a_2 B^2) r_l = 0,$$

where B is called the back-shift operator such that $Br_l = r_{l-1}$. This difference equation determines the properties of the ACF of a stationary AR(2) time series.